

Kaiden Bell

Hardware & Embedded Systems Engineering

Sparks, Nevada

(775) 412-8944 — Kaidenbell50@gmail.com — GitHub: Kaiden-Bell

Professional Profile

Computer Science and Engineering student with a strong foundation in software engineering, full-stack development, computer architecture, and systems programming. Experienced in designing and building secure backend services, modern web applications, instruction-level simulators, and embedded hardware/software systems. Passionate about building reliable, high-performance software from low-level systems through full-stack applications. Seeking software engineering internships focused on backend, full-stack, infrastructure, or systems development.

Technical Competencies

Languages

- C, C++, Python, Java, JavaScript, TypeScript, SQL, ARM Assembly

Backend & APIs

- FastAPI, Node.js
- REST API design and development
- Authentication and session management
- Serial communication (UART/pySerial)

Frontend

- React
- HTML, CSS, JavaScript, TypeScript
- Tailwind CSS

Databases

- PostgreSQL
- Relational database design

Systems Programming

- Bitwise manipulation
- Memory management
- Instruction decoding and CPU simulation
- ARMv7 architecture

Embedded Systems

- Embedded C
- State-machine design
- UART communication
- ADCs, timers, and interrupts
- Breadboard prototyping
- Hardware debugging

Security

- Argon2id

- Authenticated encryption (XChaCha20-Poly1305)
- Secure authentication workflows

Tools

- Git/GitHub
- Linux
- Docker
- AWS EC2
- Uvicorn

Machine Learning

- Feature engineering
- Model evaluation

Engineering Projects

Embedded Swamp Cooler Control System (Arduino C)

- Designed and implemented a full embedded control system using breadboarded circuitry, environmental sensors, and actuator control.
- Developed real-time control logic in C utilizing ADC readings, timers, interrupt service routines, and conditional state transitions.
- Implemented modular system architecture for maintainability and reliability.
- Performed hardware debugging to resolve sensor noise, signal stability, and integration issues.

ARMv7 Instruction-Level Emulator (C++)

- Built a functional ARMv7 emulator modeling registers, instruction decoding, execution pipeline, and memory state transitions.
- Implemented opcode parsing and execution logic for multiple instruction categories.
- Simulated program counter updates, arithmetic/logic operations, and memory access instructions.
- Designed an emulator as a learning tool for understanding assembly and processor architecture.

Secure Password Manager v1.0 (Python)

- Developed a secure credential storage system emphasizing encryption, integrity verification, and safe memory handling.
- Applied system-level security design principles and defensive programming practices.
- Focused on minimizing attack surface and improving secure memory management.

The Vault — Hardware-Gated Encrypted Password Manager v2.0 (Python, Arduino C, HTML/CSS, JavaScript)

- Developed a hardware-assisted authentication system using an Arduino Mega, 4×4 matrix keypad, and status LEDs, communicating with a Python backend over UART serial (pyserial) for secure access control.
- Designed and implemented firmware to scan keypad input, transmit authentication requests, and provide real-time hardware feedback while keeping all cryptographic operations isolated to the backend.
- Built a layered authentication architecture separating hardware access control from cryptographic key management, preventing sensitive keys or credentials from being exposed to embedded hardware.
- Engineered secure backend services using FastAPI, PostgreSQL, Argon2id, and XChaCha20-Poly1305 to manage encrypted credential storage, session state, and hardware authorization.

RLPredictor: ML Esports Data Pipeline & Prediction Engine (Python)

- Designed and implemented an end-to-end data pipeline integrating multiple external sources such as Liquipedia (web scraping), Ballchasing API (replay data), and Reddit sentiment analysis.
- Engineered a 13-feature dataset combining historical performance, head-to-head statistics, player momentum, and public sentiment for predictive modeling.
- Built and trained a PyTorch-based neural network to generate predictions for player performance metrics.
- Developed data preprocessing workflows for structured extraction, normalization, and temporal analysis across large replay datasets.
- Implemented a conversational CLI interface to translate outputs into readable explanations, combining statistical inference with logic.
- Designed a self-supervised training pipeline that generates labeled training data from historical replays, enabling continuous model improvement.

Minecraft “Random Events” Server Plugin (Java)

- Developed a modular server plugin implementing event-driven gameplay mechanics.
- Applied object-oriented programming principles and server-side state management.
- Integrated persistent score tracking and configurable gameplay logic.

Professional Experience

Independent IT & Systems Technician — Self-Employed

2025 – Present

- Diagnosed and resolved hardware, firmware, and OS-level issues across diverse client systems.
- Performed component-level troubleshooting and system configuration.
- Conducted secure system hardening and reliability optimization.

Freelance Web & Systems Developer

2024 – Present

- Designed and deployed full-stack production websites including backend APIs and database integration.
- Managed hosting environments and domain configuration.

NSHE — Circulation Senior Worker

2023 – Present

- Contributed to the efficient functioning of the Knowledge Center by assisting patrons with book-related queries.
- Enhanced communication skills while working in a collaborative and service-oriented environment.

Hidden Valley CC — Event Specialist

2021 – 2023

- Organized and executed various events.
- Led meticulous event setup, ensuring seamless event execution.
- Developed strong organizational and communication skills while contributing to the success of diverse events.

Chillworks LLC — Shift Leader

2018 – 2021

- Develop Dairy Queen’s treats in a safe, communicative environment
- Communicate with co-workers to successfully deliver dairy queen products.
- Successfully fulfill customers’ orders, needs, requests, and more.

Education

Bachelor of Science in Computer Science and Engineering

University of Nevada, Reno

Aug 2023 – May 2027 (Expected)

Coral Academy of Science HS Honors Diploma

6275 Neil Rd. Reno, NV

Aug 2019 – June 2023

Relevant Coursework

Embedded & Electrical Systems

- CPE 301 – Embedded Systems (micro-controller design, ADC, timers, interrupts, debugging)
- CPE 201 – Digital Design (logic gates, combinational/sequential circuits, breadboarding)
- EE 220 – Circuits I (nodal analysis, KCL/KVL, dependent sources, resistor networks)

Computer Architecture & Systems

- CS 219 – Computer Organization & Architecture (assembly, Boolean algebra, bitwise ops)
- CS 326 – Programming Language Concepts (syntax, grammars, Prolog)
- CS 456 – Automata and Formal Languages (Advanced syntaxes, and grammars)

Algorithms & Theory

- CS 365 – Discrete Mathematics
- CS 477 – Analysis of Algorithms
- CS 302 – Data Structures and Algorithms

Mathematics & Physics

- MATH 181 – Calculus 1
- MATH 182 – Calculus 2
- MATH 283 – Calculus 3
- MATH 285 – Differential Equations
- MATH 330 – Linear Algebra
- STAT 352 – Probability and Statistics
- PHYS 180 – Physics for Scientists and Engineers I
- PHYS 181 – Physics for Scientists and Engineers II

Data & Intelligent Systems

- CS 457 – Database Management Systems
- CS 420 – Computer-Human Interaction
- CS 422 – Introduction to Machine Learning

Engineering Communication & Leadership

- ENGR 100 – Soldering, technical communication, leadership fundamentals
- ENGR 301 – Team-based project management and engineering communication

Languages

English (Proficient)

Japanese (Intermediate)